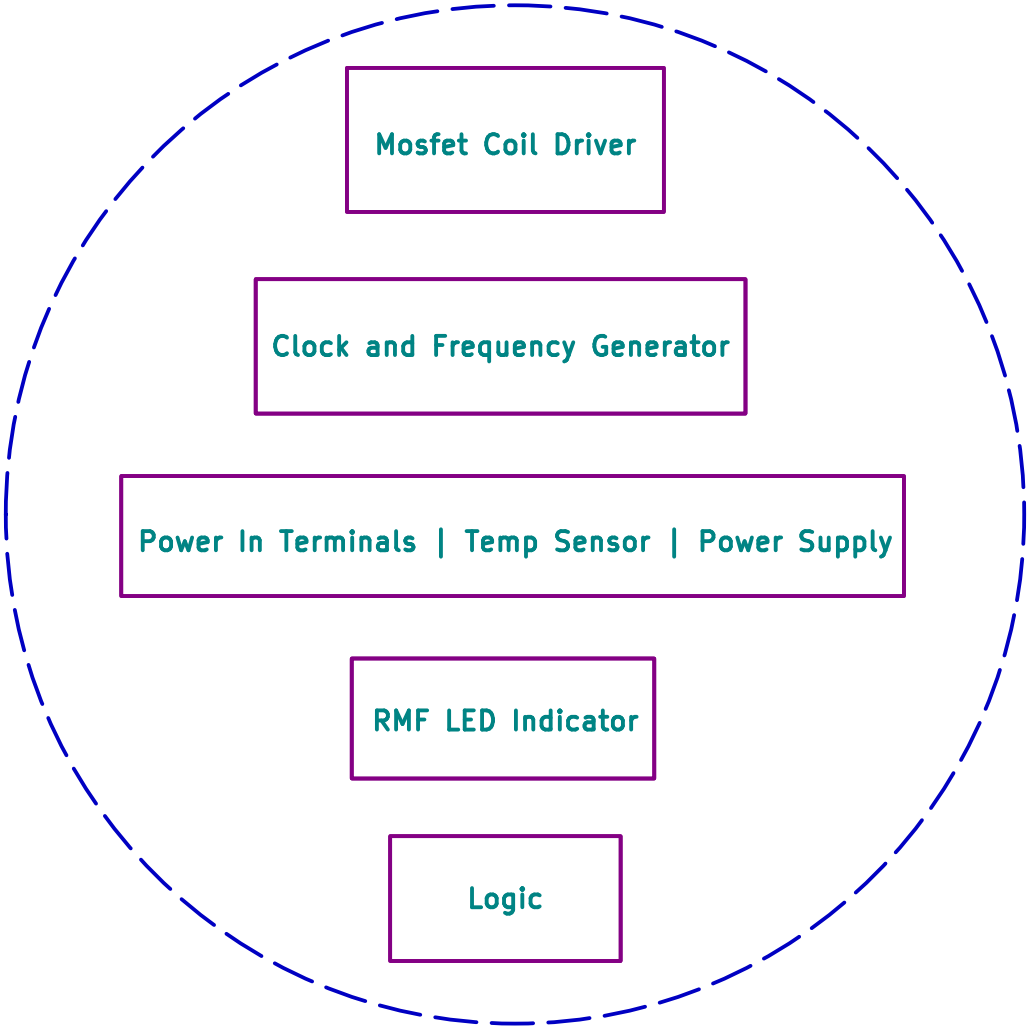


ToDo's:
- switch to 4x D-Flip Flop wit Q and $\overline{Q}$ -> done
- use fixed quarz oszi with 32 kHz -> done
- use 4x D-Flip to devide down -> done
- exchange to 4x SPDT switch to change clock and DIR -> done
- add Silk Screen Labels for Qs and $\overline{Q}$s on each IC pin
- add header and extra optocoupler for Tacho Out with V_{IN} -> w Signal, from active clock signal
- add two Ground Posts for the Osci Clip -> done
- adjust via hole from Posts to tight fit
- seperate opto supply to buffered DC/DC In Voltage
- add 10-20 22uF Cs to this supply
- exchange MosFet driver with dual ones in BGA package
- switch two Mosfets of the same kind at once to save more space
- reduce to one pcb, idealy only on top side

DC/DC::
- adjust undervoltage lockouts to 4V
- adjust L to 10µH
- change Vout to 1.8V
  -> change Rv of Opto LEDs to 51R
- remove Elko and place big ceramics
- exchange power supply to 5V LDO and dual voltage regulator for negative supply -3.3V und +3.3V
- 5V from LDO supplys VCCI on MosFet drivers
- buffered 12V Voltage supplys VCCA/VCCB of MosFet drivers LOGIC:
- checking if AND can be replaced with NANDs -> not possible/available
- add TVS Diodes on the Flip Flops Inputs
- series resistors for all outputs
- check if Ground is missing on VDDI input side GND1
- exchange SD103ASDM-7-F or rather remove it from the VDDA Path
- create a solder Jumper that is soldered during reflow and put it between Power Supplys, instead or Zero Ohms

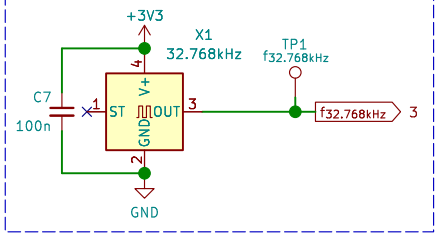
Rotating Magnetic Field Driver

Sheet: /
File: RMF_Driver.kicad_sch

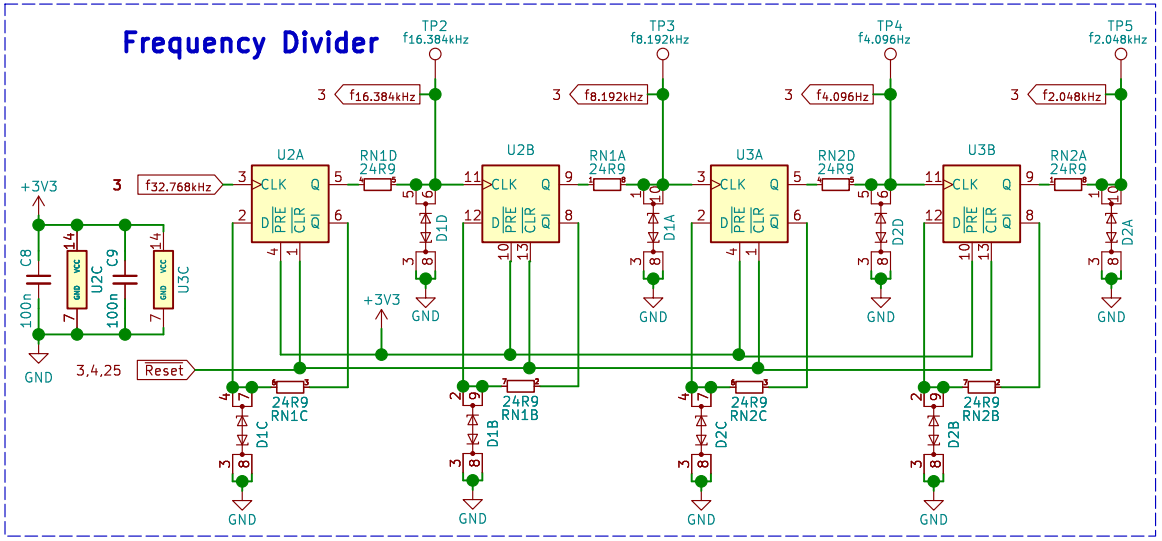
Title: RMF Driver

Size: A4	Date: 2023-01-20	**Rev: 2**
KiCad E.D.A. 9.0.6		Id: 1/33

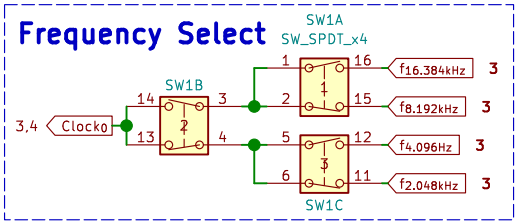
Main Frequency 32.768kHz



Frequency Divider

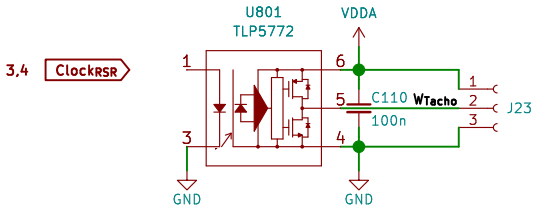


Frequency Select



Tacho Out





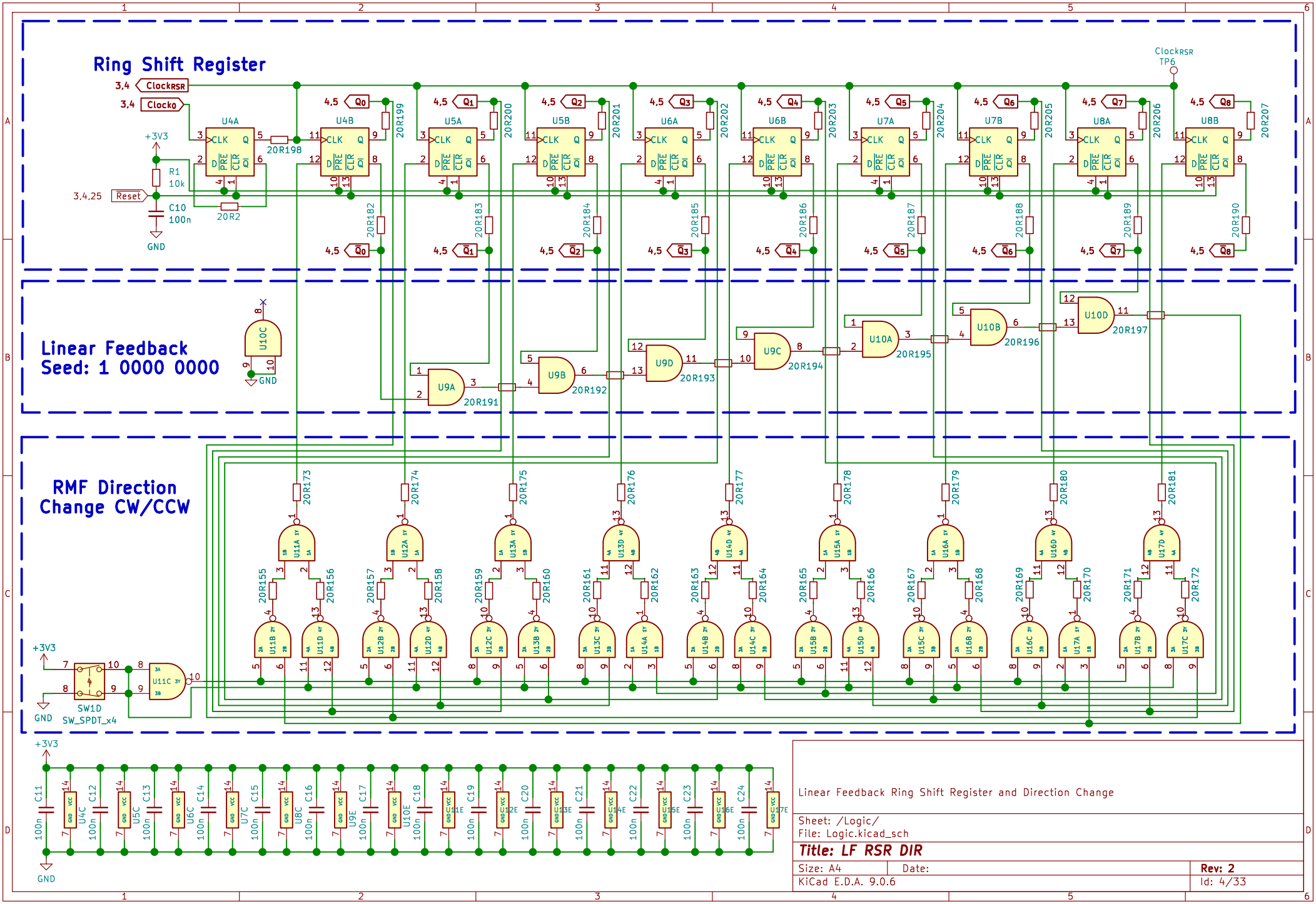
fMain[Hz]	Clock0[Hz]	ClockRSR[Hz]	vAlternator[rpm]
32768	16384	8192	9102
	8192	4096	4551
	4096	2048	2275
	2048	1024	1137

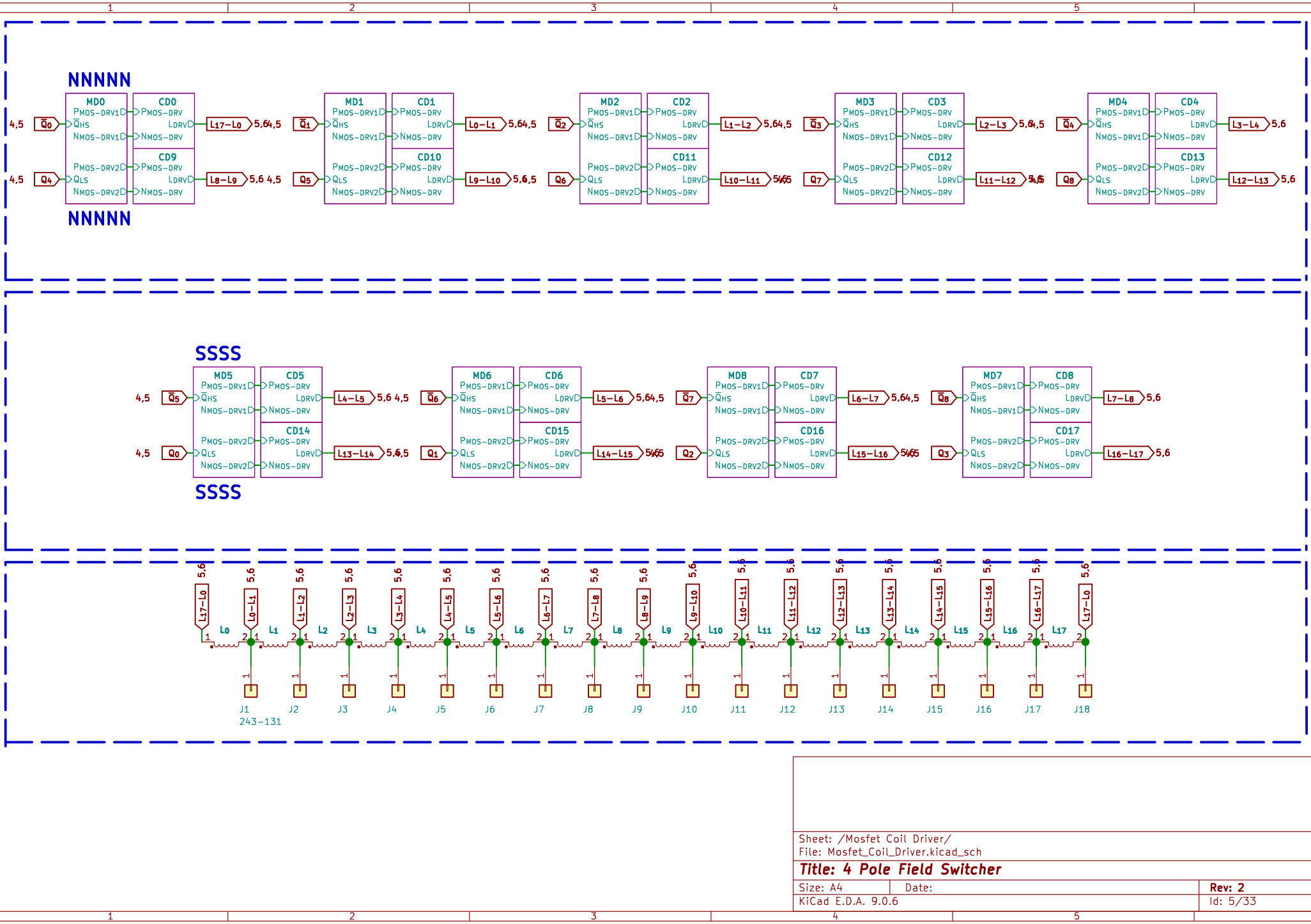
Clock and Frequency Generator

Sheet: /Clock and Frequency Generator/
File: Clock_and_Frequency_Generator.kicad_sch

Title: Clock | Frequency Generator

Size: A4	Date:	Rev: 2
KiCad E.D.A. 9.0.6		Id: 3/33

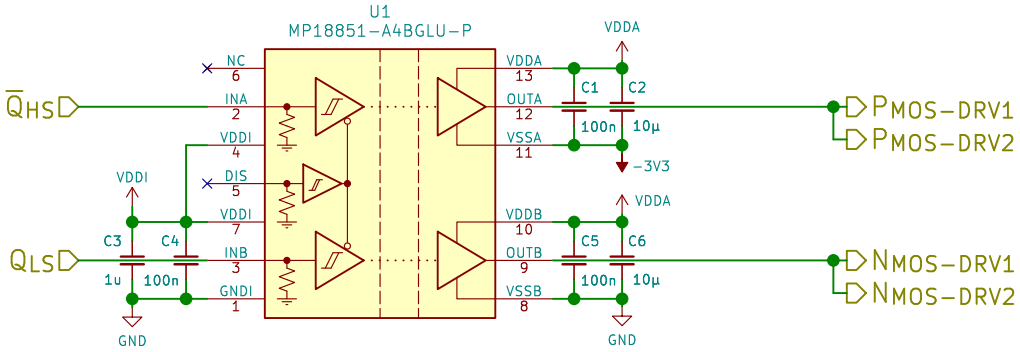

Ring Shift Register
Linear Feedback
Seed: 1 0000 0000
RMF Direction
Change CW/CCW
ClockRSR
Clock0
Reset
+3V3
GND
R1
10k
C10
100n
20R2
20R198
U4A
U4B
U5A
U5B
U6A
U6B
U7A
U7B
U8A
U8B
Q0
Q1
Q2
Q3
Q4
Q5
Q6
Q7
Q8
20R182
20R183
20R184
20R185
20R186
20R187
20R188
20R189
20R190
U10C
U9A
U9B
U9D
U9C
U10A
U10B
U10D
20R191
20R192
20R193
20R194
20R195
20R196
20R197
SW1D
SW_SPDT_x4
U11C
20R155
20R156
20R157
20R158
20R159
20R160
20R161
20R162
20R163
20R164
20R165
20R166
20R167
20R168
20R169
20R170
20R171
20R172
20R173
20R174
20R175
20R176
20R177
20R178
20R179
20R180
20R181
C11
C12
C13
C14
C15
C16
C17
C18
C19
C20
C21
C22
C23
C24
U4C
U5C
U6C
U7C
U8C
U9E
U10E
U11E
U12E
U13E
U14E
U15E
U16E
U17E
Linear Feedback Ring Shift Register and Direction Change
Sheet: /Logic/
File: Logic.kicad_sch
Title: LF RSR DIR
Size: A4
Date:
Rev: 2
KiCad E.D.A. 9.0.6
Id: 4/33



Sheet: /Mosfet Coil Driver/
File: Mosfet_Coil_Driver.kicad_sch

Title: 4 Pole Field Switcher

Size: A4	Date:	Rev: 2
KiCad E.D.A. 9.0.6		Id: 5/33

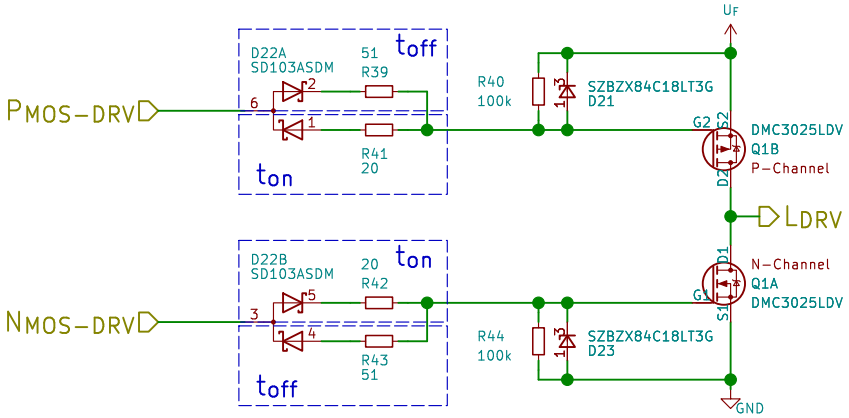


MosFet Driver

Sheet: /Mosfet Coil Driver/MD0/
File: MD.kicad_sch

Title: MD

Size: A4	Date:	**Rev: 2**
KiCad E.D.A. 9.0.6		Id: 2/33

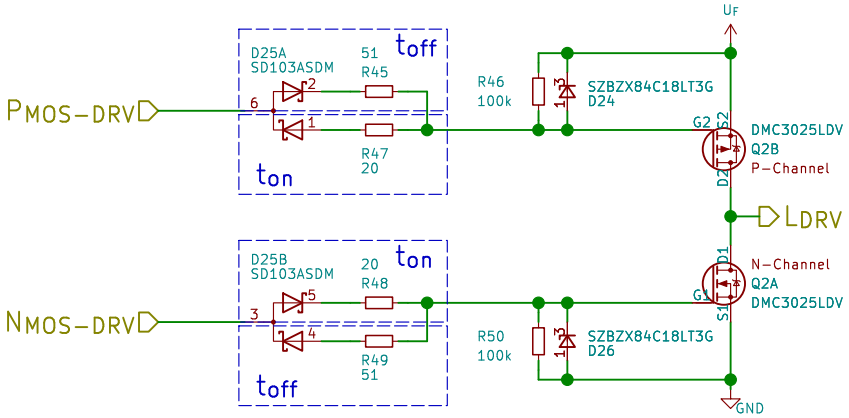


Coil Driver

Sheet: /Mosfet Coil Driver/CD1/
File: CD.kicad_sch

Title: CD

Size: A4	Date:	**Rev: 2**
KiCad E.D.A. 9.0.6		Id: 7/33

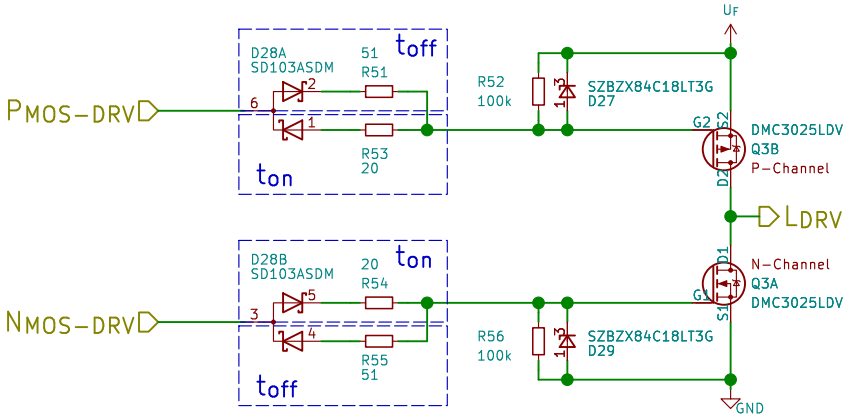


Coil Driver

Sheet: /Mosfet Coil Driver/CD0/
File: CD.kicad_sch

Title: CD

Size: A4	Date:	**Rev: 2**
KiCad E.D.A. 9.0.6		Id: 8/33

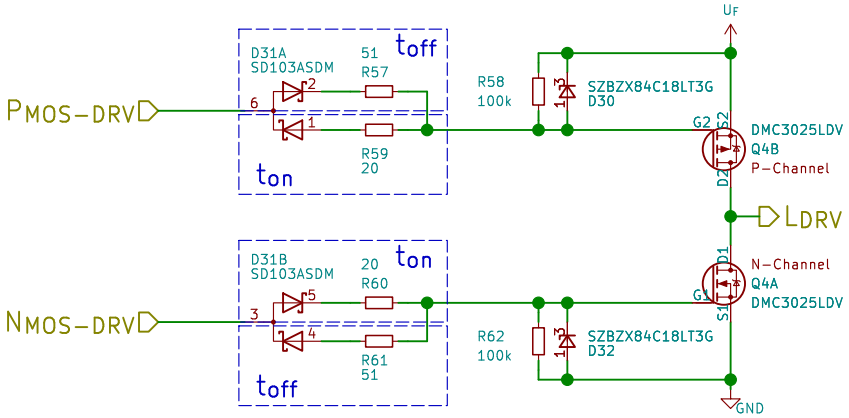


Coil Driver

Sheet: /Mosfet Coil Driver/CD2/
File: CD.kicad_sch

Title: CD

Size: A4	Date:	**Rev: 2**
KiCad E.D.A. 9.0.6		Id: 9/33

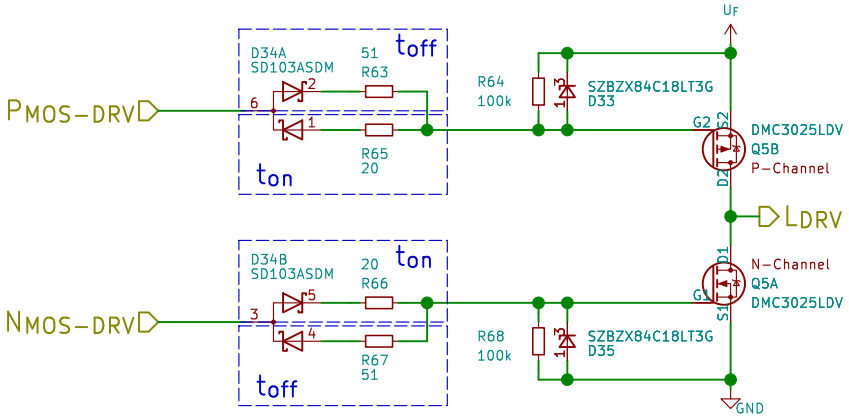


Coil Driver

Sheet: /Mosfet Coil Driver/CD3/
File: CD.kicad_sch

Title: CD

Size: A4	Date:	**Rev: 2**
KiCad E.D.A. 9.0.6		Id: 10/33

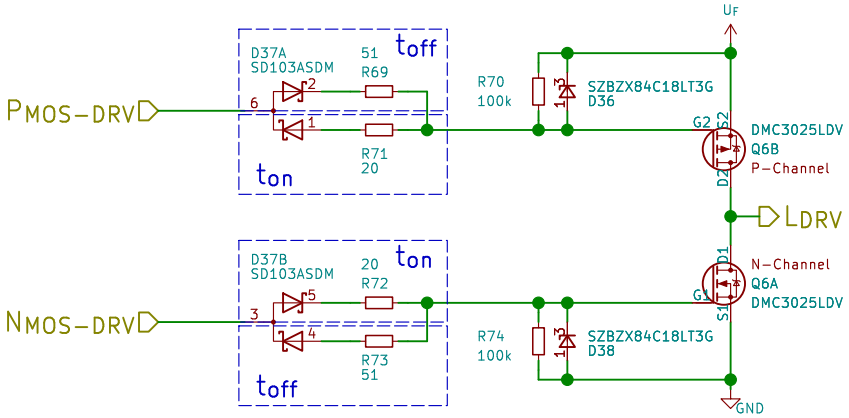


Coil Driver

Sheet: /Mosfet Coil Driver/CD4/
File: CD.kicad_sch

Title: CD

Size: A4	Date:	**Rev: 2**
KiCad E.D.A. 9.0.6		Id: 11/33

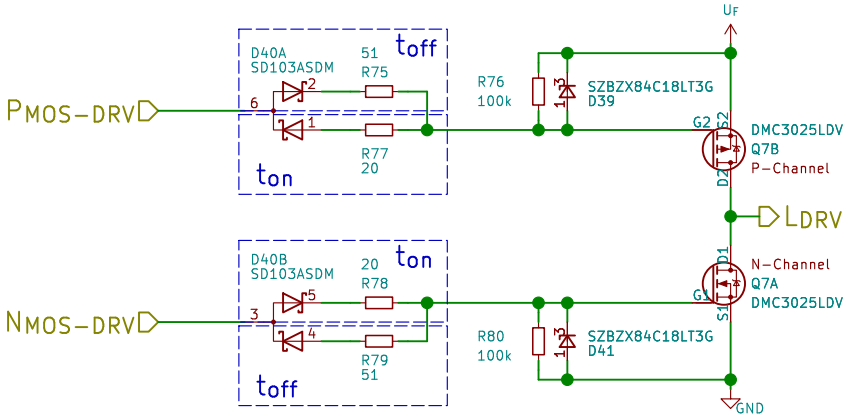


Coil Driver

Sheet: /Mosfet Coil Driver/CD5/
File: CD.kicad_sch

Title: CD

Size: A4	Date:	**Rev: 2**
KiCad E.D.A. 9.0.6		Id: 12/33

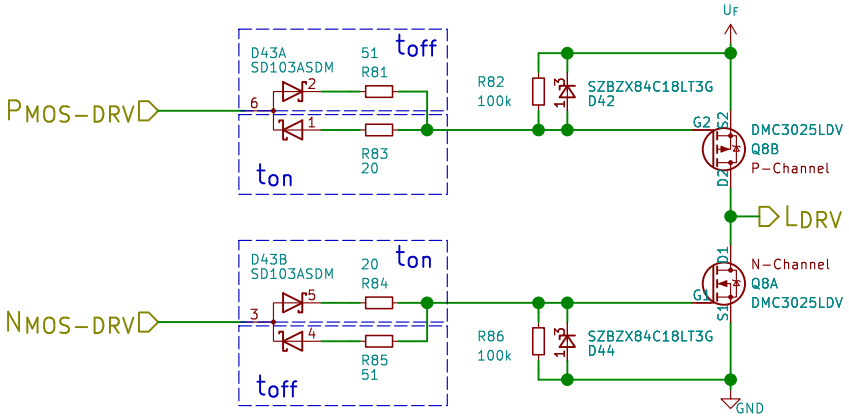


Coil Driver

Sheet: /Mosfet Coil Driver/CD6/
File: CD.kicad_sch

Title: CD

Size: A4	Date:	**Rev: 2**
KiCad E.D.A. 9.0.6		Id: 13/33

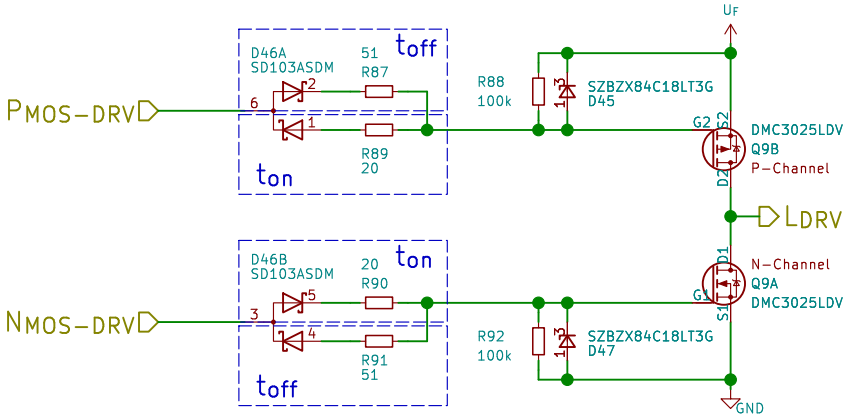


Coil Driver

Sheet: /Mosfet Coil Driver/CD7/
File: CD.kicad_sch

Title: CD

Size: A4	Date:	**Rev: 2**
KiCad E.D.A. 9.0.6		Id: 14/33

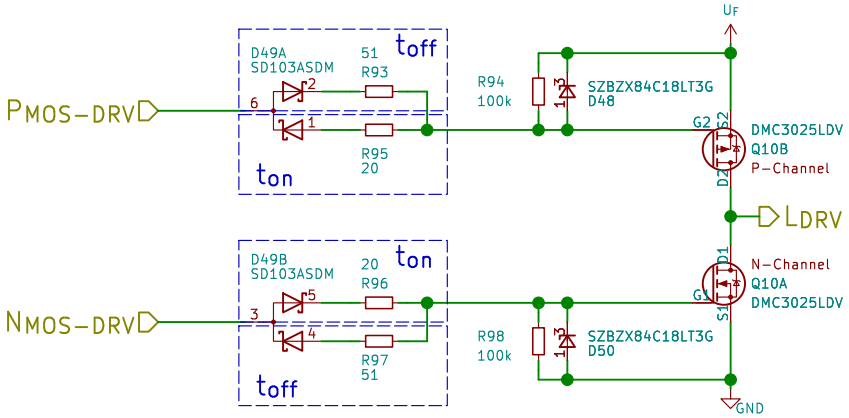


Coil Driver

Sheet: /Mosfet Coil Driver/CD8/
File: CD.kicad_sch

Title: CD

Size: A4	Date:	**Rev: 2**
KiCad E.D.A. 9.0.6		Id: 15/33

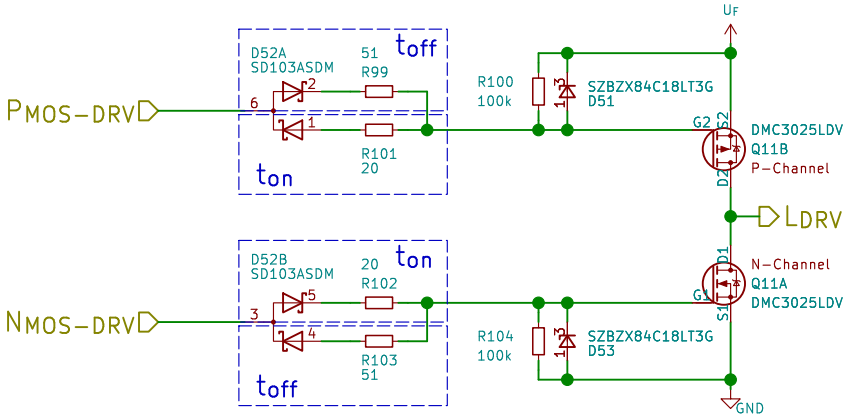


Coil Driver

Sheet: /Mosfet Coil Driver/CD11/
File: CD.kicad_sch

Title: CD

Size: A4	Date:	**Rev: 2**
KiCad E.D.A. 9.0.6		Id: 16/33

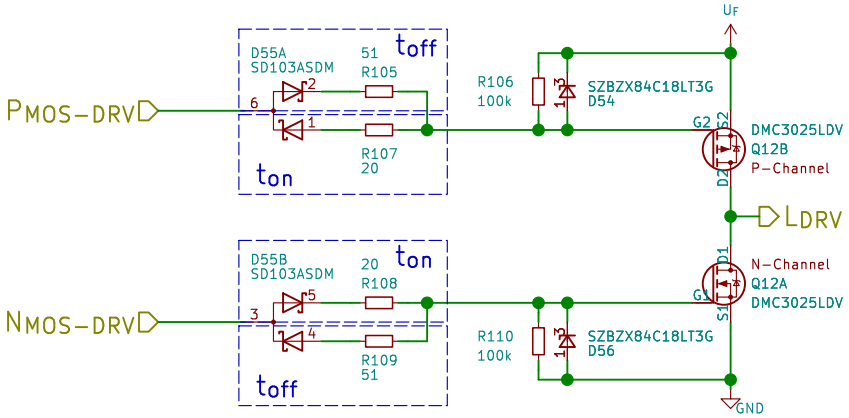


Coil Driver

Sheet: /Mosfet Coil Driver/CD9/
File: CD.kicad_sch

Title: CD

Size: A4	Date:	**Rev: 2**
KiCad E.D.A. 9.0.6		Id: 17/33

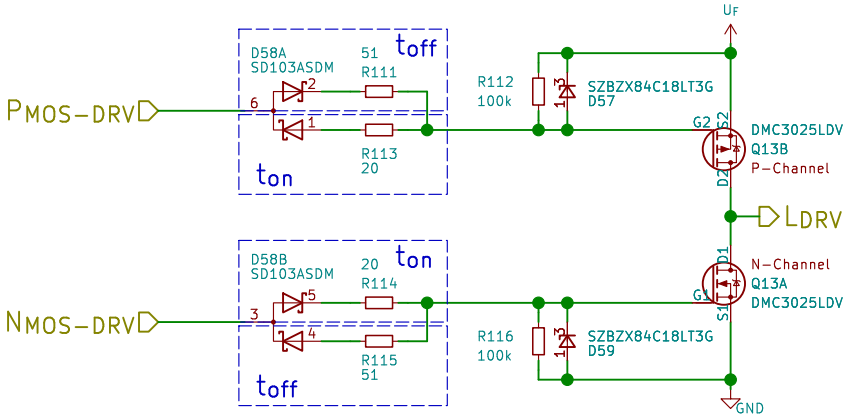


Coil Driver

Sheet: /Mosfet Coil Driver/CD10/
File: CD.kicad_sch

Title: CD

Size: A4	Date:	**Rev: 2**
KiCad E.D.A. 9.0.6		Id: 18/33

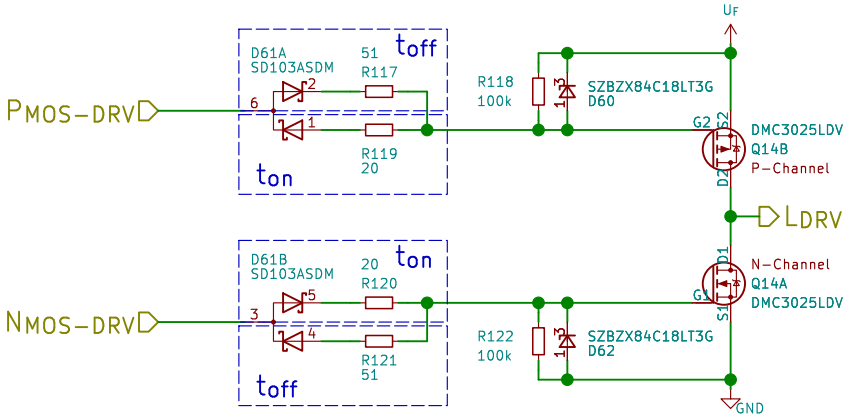


Coil Driver

Sheet: /Mosfet Coil Driver/CD16/
File: CD.kicad_sch

Title: CD

Size: A4	Date:	**Rev: 2**
KiCad E.D.A. 9.0.6		Id: 19/33

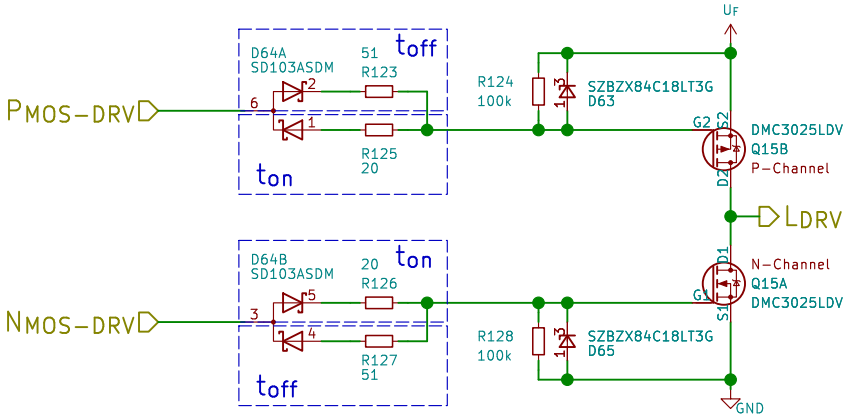


Coil Driver

Sheet: /Mosfet Coil Driver/CD13/
File: CD.kicad_sch

Title: CD

Size: A4	Date:	**Rev: 2**
KiCad E.D.A. 9.0.6		Id: 20/33

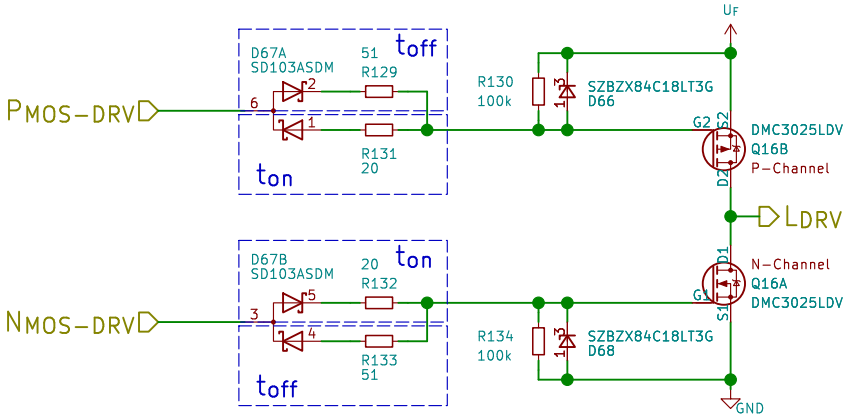


Coil Driver

Sheet: /Mosfet Coil Driver/CD17/
File: CD.kicad_sch

Title: CD

Size: A4	Date:	**Rev: 2**
KiCad E.D.A. 9.0.6		Id: 21/33

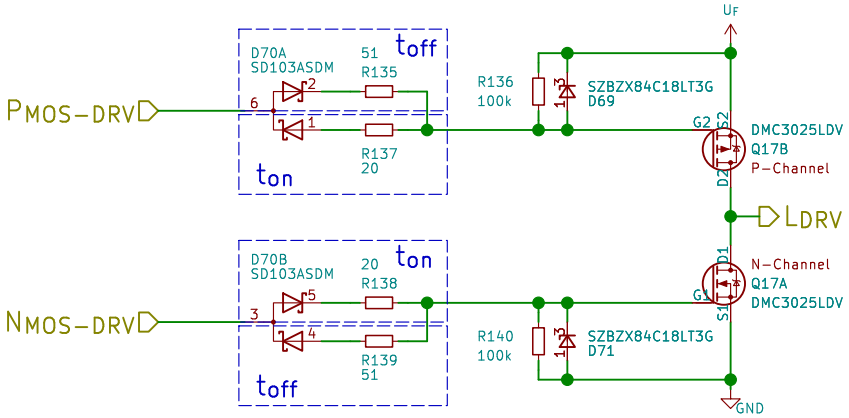


Coil Driver

Sheet: /Mosfet Coil Driver/CD14/
File: CD.kicad_sch

Title: CD

Size: A4	Date:	**Rev: 2**
KiCad E.D.A. 9.0.6		Id: 22/33

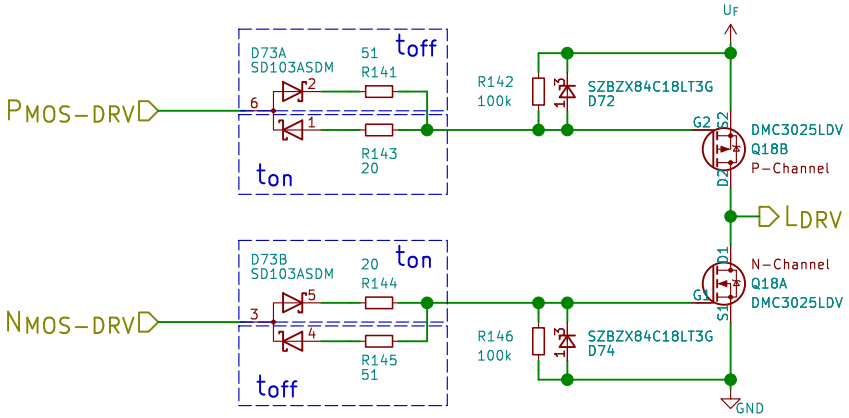


Coil Driver

Sheet: /Mosfet Coil Driver/CD15/
File: CD.kicad_sch

Title: CD

Size: A4	Date:	**Rev: 2**
KiCad E.D.A. 9.0.6		Id: 23/33

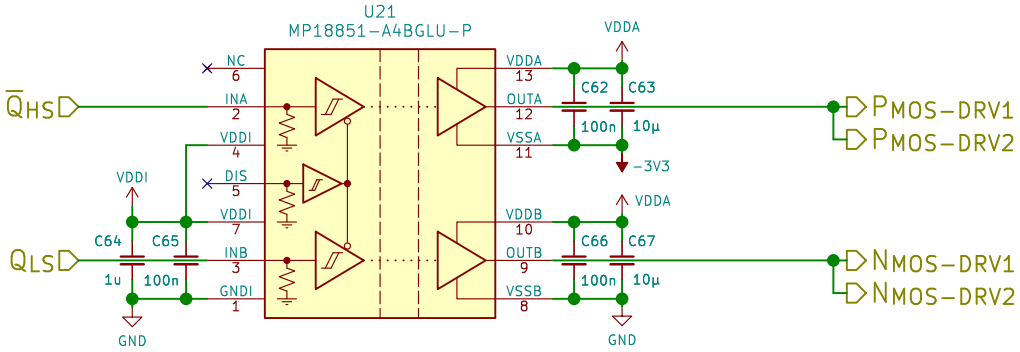


Coil Driver

Sheet: /Mosfet Coil Driver/CD12/
File: CD.kicad_sch

Title: CD

Size: A4	Date:	**Rev: 2**
KiCad E.D.A. 9.0.6		Id: 24/33

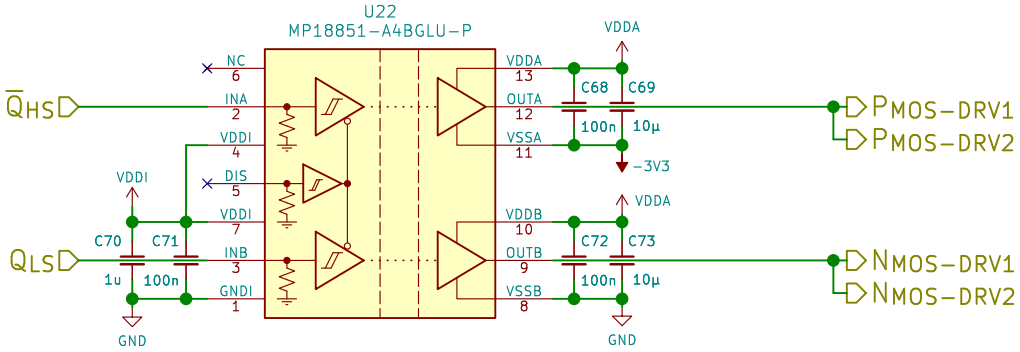


MosFet Driver

Sheet: /Mosfet Coil Driver/MD1/
File: MD.kicad_sch

Title: MD

Size: A4	Date:	**Rev: 2**
KiCad E.D.A. 9.0.6		Id: 26/33

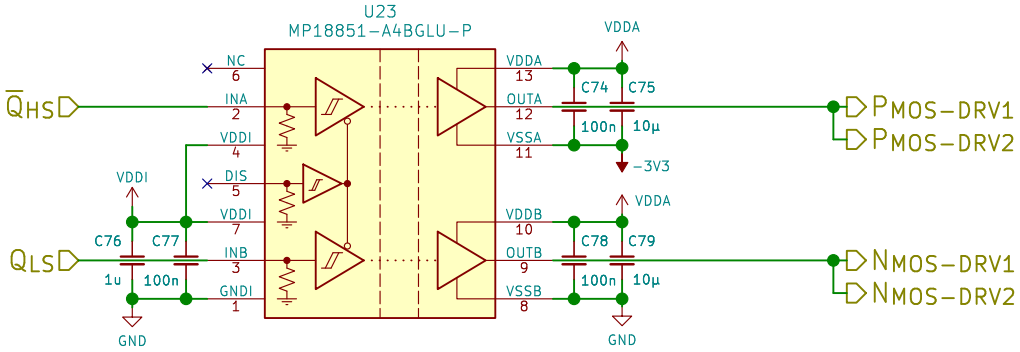


MosFet Driver

Sheet: /Mosfet Coil Driver/MD5/
File: MD.kicad_sch

Title: MD

Size: A4	Date:	**Rev: 2**
KiCad E.D.A. 9.0.6		Id: 27/33

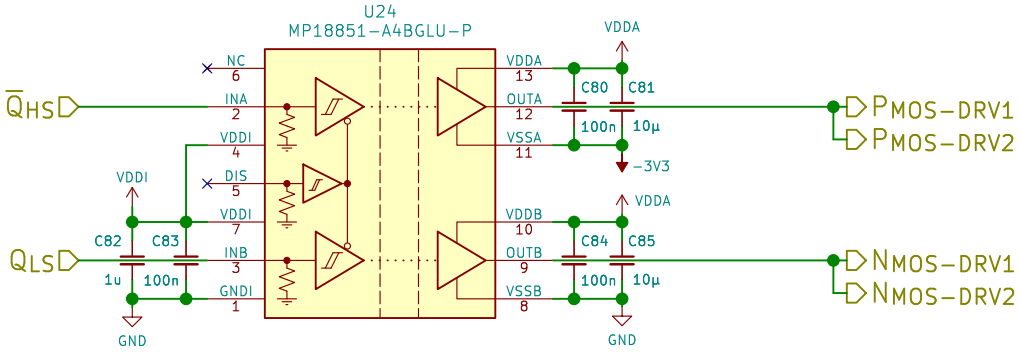


MosFet Driver

Sheet: /Mosfet Coil Driver/MD2/
File: MD.kicad_sch

Title: MD

Size: A4	Date:	**Rev: 2**
KiCad E.D.A. 9.0.6		Id: 29/33

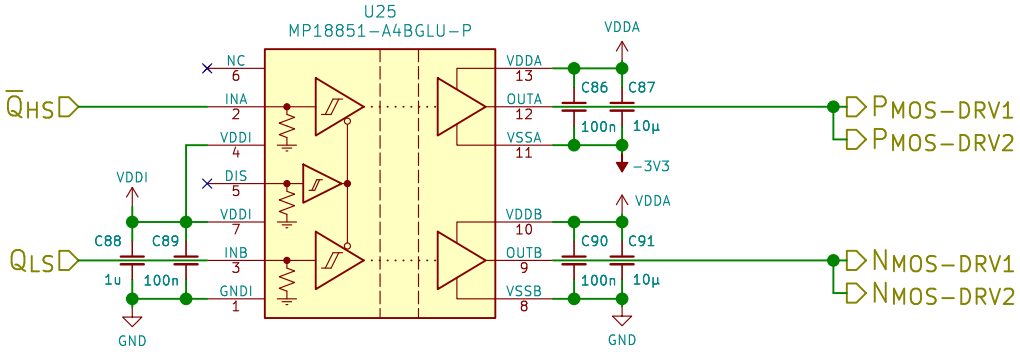


MosFet Driver

Sheet: /Mosfet Coil Driver/MD3/
File: MD.kicad_sch

Title: MD

Size: A4	Date:	**Rev: 2**
KiCad E.D.A. 9.0.6		Id: 32/33

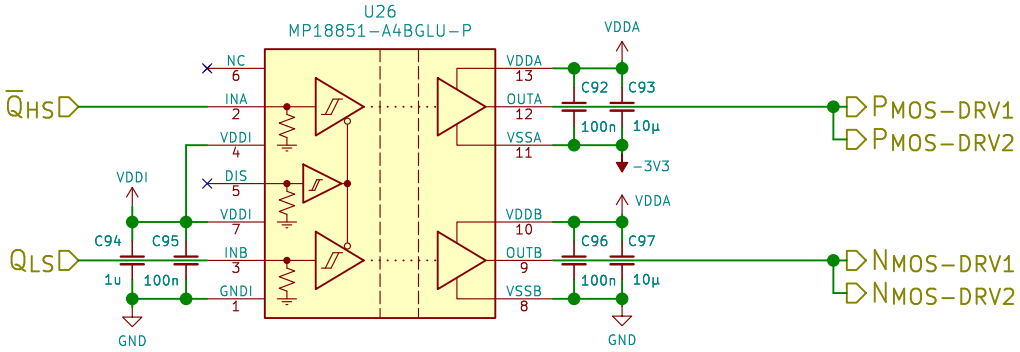


MosFet Driver

Sheet: /Mosfet Coil Driver/MD4/
File: MD.kicad_sch

Title: MD

Size: A4	Date:	**Rev: 2**
KiCad E.D.A. 9.0.6		Id: 35/33

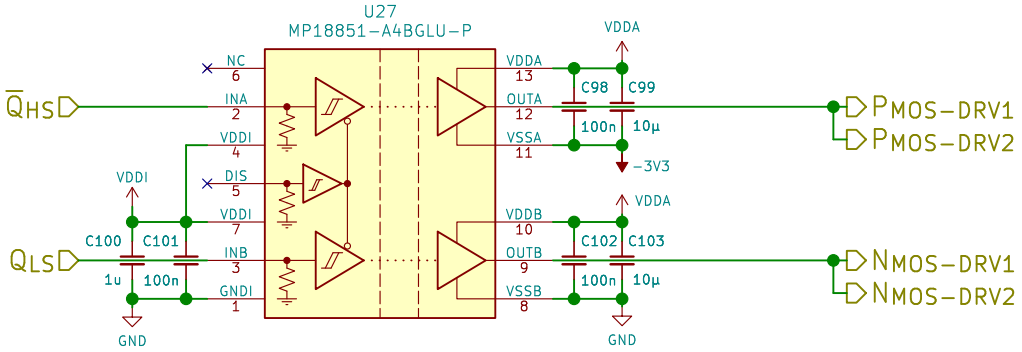


MosFet Driver

Sheet: /Mosfet Coil Driver/MD6/
File: MD.kicad_sch

Title: MD

Size: A4	Date:	**Rev: 2**
KiCad E.D.A. 9.0.6		Id: 39/33

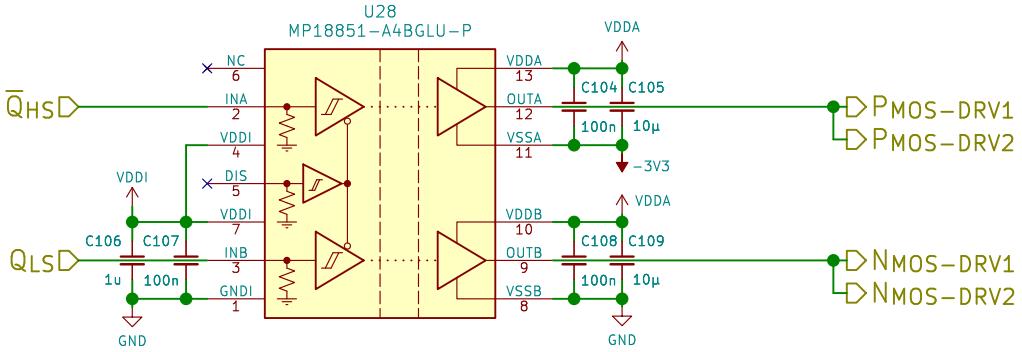


MosFet Driver

Sheet: /Mosfet Coil Driver/MD8/
File: MD.kicad_sch

Title: MD

Size: A4	Date:	**Rev: 2**
KiCad E.D.A. 9.0.6		Id: 40/33

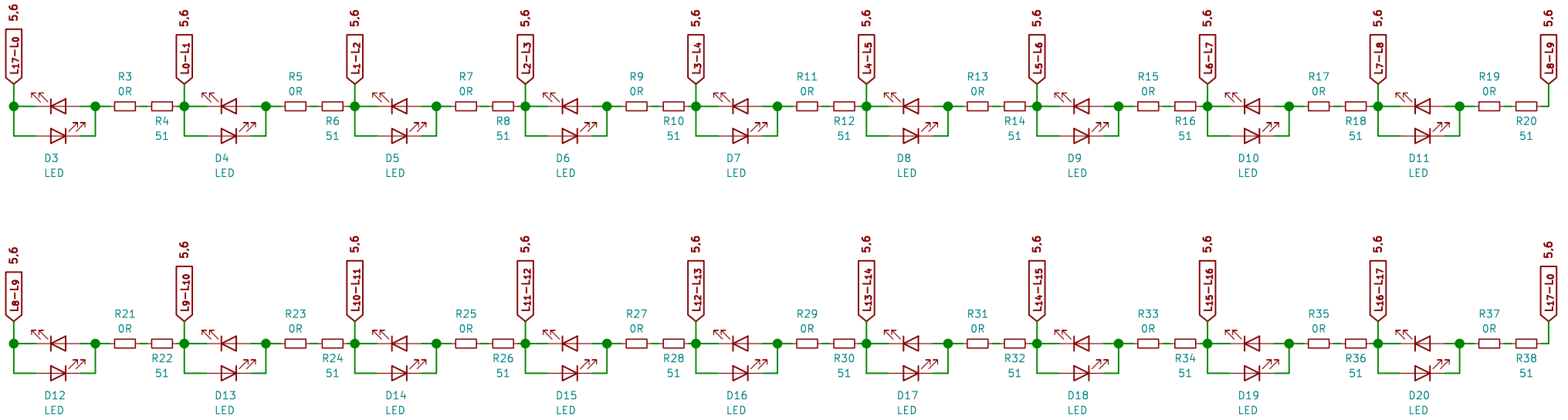


MosFet Driver

Sheet: /Mosfet Coil Driver/MD7/
File: MD.kicad_sch

Title: MD

Size: A4	Date:	**Rev: 2**
KiCad E.D.A. 9.0.6		Id: 41/33

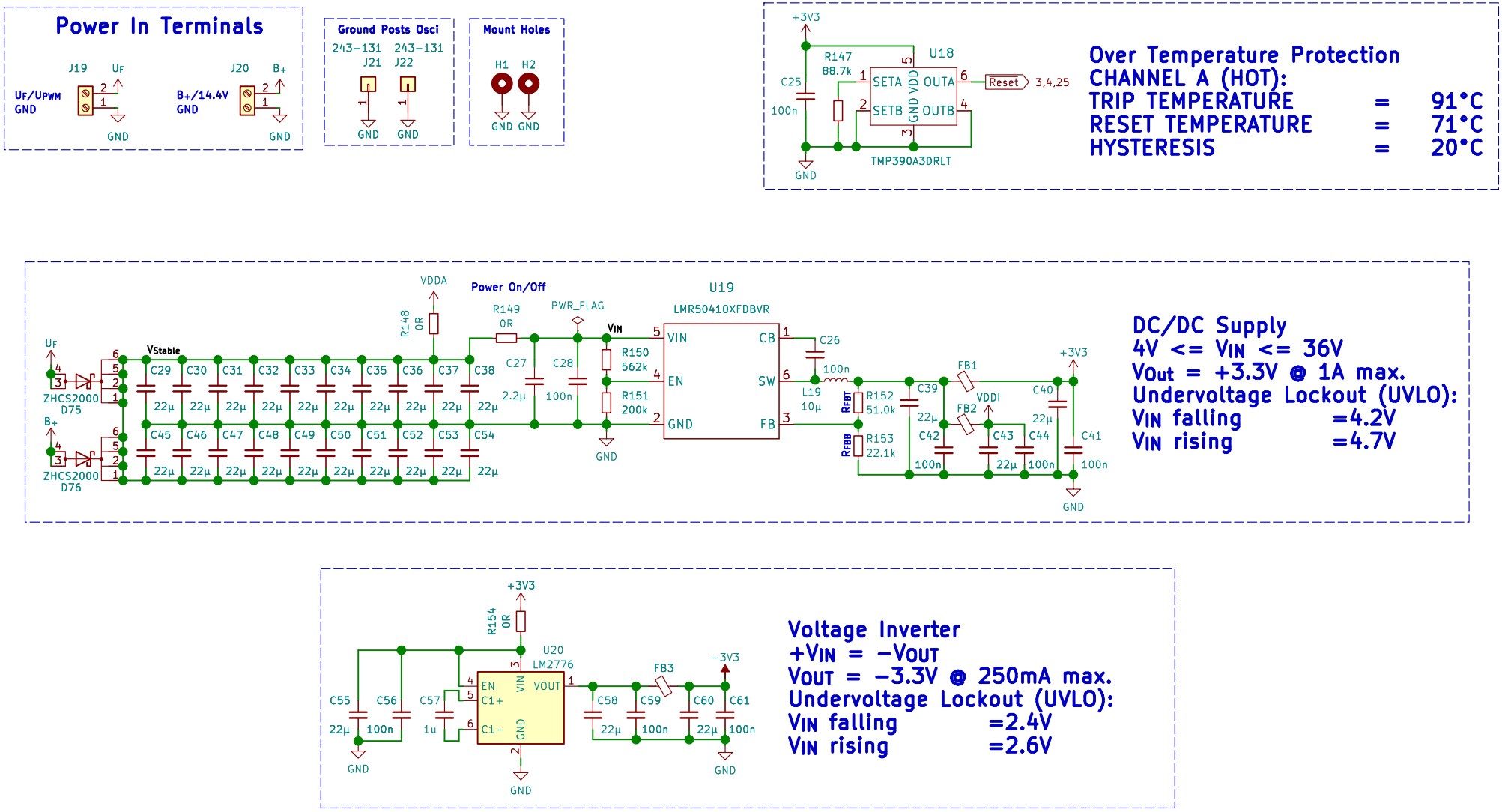


Pole Motion Direction LEDs

Sheet: /RMF LED Indicator/
File: RMF_LED_Indicator.kicad_sch

Title: LED Indicator

Size: A4	Date:	**Rev: 2**
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Power In Terminals
J19
UF
UF/UPWM
GND
GND
J20
B+
B+/14.4V
GND
GND
Ground Posts Osci
243-131
243-131
J21
J22
GND
GND
Mount Holes
H1
H2
GND
GND
+3V3
R147
88.7k
U18
C25
100n
SETA
OUTA
SETB
OUTB
VDD
GND
Reset
3,4,25
TMP390A3DRLT
GND
Over Temperature Protection
CHANNEL A (HOT):
TRIP TEMPERATURE = 91°C
RESET TEMPERATURE = 71°C
HYSTERESIS = 20°C
UF
B+
ZHCS2000
D75
ZHCS2000
D76
VStable
C29
C30
C31
C32
C33
C34
C35
C36
C37
C38
C45
C46
C47
C48
C49
C50
C51
C52
C53
C54
22µ
VDDA
R148
0R
Power On/Off
R149
0R
PWR_FLAG
C27
2.2µ
C28
100n
VIN
R150
562k
R151
200k
GND
U19
LMR50410XFDBVR
VIN
EN
GND
CB
SW
FB
C26
100n
L19
10µ
RFBT
R152
51.0k
RFBB
R153
22.1k
C39
22µ
C42
100n
FB1
FB2
VDDI
C43
22µ
C44
100n
C40
22µ
C41
100n
+3V3
GND
DC/DC Supply
4V <= VIN <= 36V
VOut = +3.3V @ 1A max.
Undervoltage Lockout (UVLO):
VIN falling =4.2V
VIN rising =4.7V
+3V3
R154
0R
U20
LM2776
EN
VIN
VOUT
C1+
C1−
GND
C55
22µ
C56
100n
C57
1u
FB3
−3V3
C58
22µ
C59
100n
C60
22µ
C61
100n
GND
GND
Voltage Inverter
+VIN = −VOUT
VOUT = −3.3V @ 250mA max.
Undervoltage Lockout (UVLO):
VIN falling =2.4V
VIN rising =2.6V
Power In Connectors and Temperatur Sensor
UF PWR_FLAG +3V3 PWR_FLAG B+
PWR_FLAG GND PWR_FLAG −3V3 PWR_FLAG
Sheet: /Power In Terminals | Temp Sensor | Power Supply/
File: PowerInConnTempSensor.kicad_sch
Title: Power In Connectors and Temperatur Sensor
Size: A4
Date:
Rev: 2
KiCad E.D.A. 9.0.6
Id: 25/33